Research Statement

Zoë Cullen

I am a labor economist studying how opportunities are created, and how workers benefit from them. My research examines how relationships, technology and institutions affect productivity and the negotiating positions of workers and employers. A central theme is that policies intended to improve workers' outcomes can also change the incentives of employers and coworkers, with consequences that sometimes undermine their intended benefits. My work on pay transparency illustrates this tension. Pay information can trigger valuable renegotiations, but it can also incentivize employers to bargain more aggressively knowing that a concession to one worker implies concessions to those who witness it. I combine surveys, administrative data, field experiments, and theory to study these mechanisms and their implications for productivity, inequality, and worker welfare.

My research on AI extends this agenda to the value and ownership of workers' expertise. The relevant information asymmetries run in both directions: workers lack knowledge about employers and employment opportunities, while employers lack access to some of the knowledge their workers possess. Workers supply both direct labor, through the tasks they perform, and indirect labor, through the data their work generates and that can train models to perform similar tasks. Sharing this knowledge can increase productivity while changing workers' future bargaining positions. My research asks what determines the supply and value of this expertise, and what institutions allow workers to share in the returns it generates.

The projects below develop this agenda through four connected questions. How does bargaining and access to information shape wages and career decisions; what determines the supply and ownership of knowledge; how does learning about competitors affect wages and technology investment; and how do institutions and social relationships shape access to work. Across these settings, I study both the forces that create economic opportunities and the equilibrium responses that determine who ultimately gains.

Information, bargaining, and the returns to work

Pay transparency provides a useful starting point because it changes what workers know without directly regulating their wages. Yet its effects depend on whose pay is revealed and how others respond. In **Equilibrium Effects of Pay Transparency** (*Econometrica*, 2023), with Bobak Pakzad-Hurson, I study bargaining under two-sided incomplete information. When coworkers can observe one another's wages, a raise to one worker reveals information about the employer's willingness to pay and can prompt others to renegotiate. Employers consequently have a stronger incentive to resist individual demands. Transparency can therefore compress wages while lowering their average level. An event-study analysis of U.S. state laws protecting workers' right to discuss pay finds an approximately 2% wage decline, with smaller declines where workers have less scope for individual bargaining. The result illustrates why giving workers more information need not increase their share of surplus, and distinguishes transparency among coworkers from information about outside opportunities.

Information also affects workers' willingness to bargain in the first place. In **Pushing the Envelope: The Effects of Salary Negotiations** (revise and resubmit, *Econometrica*), with Bobak Pakzad-Hurson and Ricardo Perez-Truglia, I study job seekers who already have access to detailed compensation data through Levels.fyi. Even in this relatively well-informed labor market, workers face uncertainty about whether employers welcome negotiation and fear that asking for more could jeopardize an offer. In field experiments involving more than 3,100 U.S. technology-sector job seekers, light-touch encouragement

increases negotiation attempts and the probability of obtaining better compensation. A substantial discount on professional coaching, by contrast, does not significantly increase negotiation attempts on average. Women respond more strongly to the interventions. A model incorporating risk and information frictions explains why workers may decline to negotiate despite potential gains, and how encouraging negotiation can affect both the division of surplus and the allocation of workers across employers. Knowing what other jobs pay is not the same as knowing how safely one can act on that information.

Workers also use pay information to evaluate future opportunities. In **How Much Does Your Boss Make? The Effects of Salary Comparisons** (*Journal of Political Economy*, 2022), with Ricardo Perez-Truglia, I combine an information experiment with personnel and performance records at a large financial institution. Employees substantially underestimate managerial pay. Learning that managers earn more than expected increases effort and output, whereas learning that peers earn more reduces effort. The contrast is consistent with higher-level salaries conveying information about career returns, while peer salaries invite unfavorable comparisons. It highlights the importance of distinguishing information about opportunities from information about relative standing. My article **Is Pay Transparency Good?** (*Journal of Economic Perspectives*, 2024) brings together evidence on these distinctions, separating horizontal, vertical, and cross-employer transparency rather than treating disclosure as a single policy.

The relationship between information and bargaining also depends on whether workers negotiate individually or collectively. In **How Does Wage Inequality Affect the Labor Movement?** (accepted, *Quarterly Journal of Economics*), with Barbara Biasi, Julia Gilman, and Nina Roussille, I combine a vignette experiment with union organizers, a policy-induced change in wage inequality among Wisconsin teachers, and an information intervention during the 2023 Writers Guild of America strike. Greater inequality weakens support for collective bargaining, particularly among workers with stronger individual bargaining positions. Organizers responding to hypothetical choices also shift toward smaller, more homogeneous bargaining units and away from wage-centered campaigns. These findings suggest a feedback mechanism through which inequality can weaken institutions that would otherwise counteract it. They also motivate a question running through my AI research: when individual transactions impose costs on other workers, what sustains collective bargaining over the resulting surplus?

The supply and ownership of knowledge

Information frictions are partly the outcome of deliberate decisions about what to reveal. In **The Salary Taboo: Privacy Norms and the Diffusion of Information** (*Journal of Public Economics*, 2023), with Ricardo Perez-Truglia, I use a field experiment with 755 employees to study why valuable salary information does not circulate freely. Many workers are unwilling to reveal their pay and reluctant to ask colleagues about theirs, even when offered financial incentives. Formal permission to discuss pay therefore need not overcome the social costs of doing so. In **Demand for Privacy from the U.S. Government: Evidence from the Census Income Question** (revise and resubmit, *Journal of Public Economics*), with Tom Nicholas, I examine income withholding in the 1940 Census. The evidence links nonresponse to concerns about identifiable disclosure and suggests that privacy-sensitive reporting can distort the measurement of inequality. Together, these papers study how the anticipated uses of information affect its supply.

My research on AI extends this question from information about earnings to the knowledge that generates them. In **Labor as Capital: AI and the Ownership of Expertise**, with Danielle Li and Shengwu Li, I combine a survey of 4,275 full-time U.S. workers with a field experiment spanning 61 consulting markets. Workers report substantial expertise that their employers have not captured in a form usable

for AI training. In the field experiment, stronger career incentives lead employees to contribute more documentation, work examples, and sensitive knowledge to the development of AI coworkers. The availability of training data is therefore not simply a technological constraint: it responds to workers' incentives.

The theory is broader than a particular form of workplace monitoring. Today's work produces current output and can also generate knowledge that remains productive without the worker's continued participation. In our dynamic model, transferring that knowledge can improve an employer's future outside option. With incomplete contracting and limited commitment, workers may withhold knowledge to protect their future bargaining position. Individual data ownership can counter withholding, but it leaves a competition externality: one worker's contribution can weaken another's negotiating position. Collective governance of data can address this externality without requiring collective wage bargaining. The analysis identifies conditions under which workers can share knowledge efficiently and participate in the gains from AI, while distinguishing compensation for current labor from returns to the knowledge it creates.

In a law review article, **Knowledge Guilds: Sharing the Productivity Gains of AI**, with Salomé Viljoen and Danielle Li, I examine institutional arrangements for bargaining over these returns. The article develops a proposal for collective governance of work-generated knowledge, bringing together control over its use, privacy interests, and claims on the value it creates. This connects my work on the incentives to disclose information with my research on the strengths and limits of collective bargaining. The question is not only whether workers should share more knowledge, but what arrangements make sharing compatible with their longer-run interests.

The same distinction between supplying labor and supplying a reusable stock of knowledge motivates **A Market for Human Expertise: Observations**, with Alberto Cavallo and Andrew Koh. Using daily observations of a platform on which AI laboratories recruit experts, we study demand for expertise as training data. Hiring proceeds in successive waves across domains, and posted pay falls over the observation window in several domains as hiring subsides, though not uniformly. We interpret these patterns with a model in which accumulated training data face diminishing returns within a domain but can raise the value of expertise in complementary domains. The short-run evidence is consistent with a declining marginal value of additional expertise in some areas even as demand opens elsewhere. This work asks how the market for knowledge changes when buyers accumulate an asset rather than repeatedly purchase the labor needed to perform a task.

Learning about competitors: wages and technology

Information frictions on the employer side are equally consequential for labor-market outcomes. In **What's My Employee Worth? The Effects of Salary Benchmarking** (*Review of Economic Studies*, 2026), with Shengwu Li and Ricardo Perez-Truglia, I study access to market salary information using payroll records and data on the rollout and use of ADP's benchmarking tool. Access reduces new-hire salary dispersion around occupation-specific benchmarks by about 25%. The average wage effect is not statistically distinguishable from zero, but wages rise in lower-skill positions, with suggestive evidence of improved retention. The findings identify uncertainty about market wages as a source of pay dispersion. Our model shows how employers with correlated private valuations can offer different wages to identical workers because they hold different beliefs about competing offers. Public information can reduce information rents and sharpen competition. Thus, greater employer access to information need not increase employer bargaining power, just as greater worker access need not increase worker bargaining power.

This work leads naturally to another margin of labor demand: investment in technology. Employers may lack information not only about competitors' wages, but also about their adoption of AI and other technologies. In **The Innovation Race: Experimental Evidence on Advanced Technologies**, with Ester Faia, Elisa Guglielminetti, Ricardo Perez-Truglia, and Concetta Rondinelli, I study an information experiment embedded in a Bank of Italy survey of approximately 3,000 businesses. Respondents generally underestimate adoption among comparable businesses. Providing information changes their expectations about competitors and increases their own intended adoption of robotics. We do not detect a statistically significant effect on intended AI adoption. The findings provide evidence of strategic complementarities in technology investment and show that correcting beliefs can influence plans even without changing financing or access to the technology itself.

In **Innovation Without Borders? The Geography of Technological Diffusion**, with Ursel Baumann, Ester Faia, Annalisa Ferrando, Ricardo Perez-Truglia, and Judit Rariga, I extend this approach through a collaboration with the European Central Bank. A field experiment covering 3,300 businesses in twelve EU countries provides information about domestic and foreign AI investment. Respondents underestimate both, but their investment intentions respond much more strongly to expected investment by domestic peers. We find a statistically significant response to domestic investment expectations and no statistically significant response to foreign expectations. The geographic reach of learning and strategic interaction therefore matters for technology diffusion.

Taken together, the benchmarking and technology projects ask whether unequal access to market information contributes to differences in pay and investment decisions. Benchmarking provides evidence of wage convergence; whether correcting technology-investment beliefs similarly narrows adoption gaps remains a question for subsequent work. Connecting investment intentions to realized adoption, employment, and skill demand is an important next step. These projects complement the research on knowledge supply: understanding the labor-market effects of AI requires explaining both what induces employers to invest and what induces workers to supply the expertise that makes those investments useful.

Related projects examine information closer to the point of hiring and pay setting. Work with ADP on internal pay and retention data studies whether making disparities visible changes compensation and representation, and whether concerns about liability discourage information gathering. In "Labor Shortages and Firm Search," with Hoffman and Koenig, I examine whether wage-setting tools help employers learn about labor supply and respond to recruitment difficulties. These projects extend the agenda from access to information to the practical ability and willingness to use it.

Access to work and labor-market adjustment

Information also shapes who is hired and promoted, not only the wages negotiated conditional on employment. In **The Old Boys' Club: Schmoozing and the Gender Gap** (*American Economic Review*, 2023), with Ricardo Perez-Truglia, I use quasi-random manager rotations and shared smoking habits to study social interactions at a financial institution. Greater social contact with managers improves promotion prospects without corresponding improvements in measured effort or performance. The male-to-male advantage accounts for roughly one-third of the gender gap in promotions in this setting. The study identifies a channel through which unequal access to influential relationships can shape careers even when measured productivity does not differ. The results distinguish access to productive knowledge from access to relationships that influence the allocation of career rewards.

In **Increasing Demand for Workers with a Criminal Background** (*Quarterly Journal of Economics*, 2023), with Will Dobbie and Mitchell Hoffman, I study how uncertainty about productivity and downside risk limits employment opportunities. A randomized hiring-policy experiment on a nationwide staffing platform finds that the share of businesses willing to work with applicants with a criminal record rises from 39% to more than 50% when offered crime and safety insurance, a performance review, or a background check limited to the preceding year. The experiment elicits hiring choices that determine which workers can subsequently accept jobs on the platform. The findings identify alternatives to both blanket exclusion and suppressing information: supplying more relevant information or insuring specific risks can expand access to work.

Complementary research examines how market structure and nonwage preferences mediate adjustment. In **Outsourcing Tasks Online: Matching Supply and Demand on Peer-to-Peer Internet Platforms** (*Management Science*, 2021), with Chiara Farronato, I find that highly elastic labor supply on TaskRabbit accommodates demand increases largely through additional tasks rather than higher task prices. This provides a useful contrast to markets in which information or bargaining frictions slow adjustment. In **Home Sweet Home: How Much Do Employees Value Remote Work?** (*AEA Papers and Proceedings*, 2025), with Bobak Pakzad-Hurson and Ricardo Perez-Truglia, I use actual job choices to estimate a substantial value of remote work among U.S. technology workers, without finding the corresponding wage discount that a simple compensating-differentials account would predict. **The Rise of Remote Work: Evidence from Business and Worker Surveys over the COVID-19 Pandemic** (*Journal of Economics & Management Strategy*, 2025), with Bartik, Glaeser, Luca, and Stanton, complements this evidence by documenting changing employer productivity assessments and heterogeneous worker preferences.

My pandemic research studies employment adjustment when uncertainty and financial constraints are especially acute. **The Impact of COVID-19 on Small Business Outcomes and Expectations** (*PNAS*, 2020), with Bartik, Bertrand, Glaeser, Luca, and Stanton, documents early layoffs, closures, and financial fragility. **Determinants of Small Business Reopening Decisions After COVID Restrictions Were Lifted** (*Journal of Policy Analysis and Management*, 2022), with Balla-Elliott, Glaeser, Luca, and Stanton, shows how expected demand affects reopening plans. In **When Should Public Programs Be Privately Administered? Theory and Evidence from the Paycheck Protection Program** (forthcoming, *Review of Economics and Statistics*), with Bartik, Glaeser, Luca, Stanton, and Sunderam, I study the trade-off between rapid delivery of employment-preserving aid and better targeting. Using variation in banks' processing speeds, the analysis suggests that the costs of delaying support outweighed the gains from improved targeting in this setting. These projects extend my interest in decision-relevant information to the design of policies that preserve employment relationships during shocks.

Across this research, my objective is to understand how the information available to workers and employers, and the institutions governing its use, shape labor-market opportunity and the distribution of surplus. The expansion of AI changes not only demand for labor, but also what knowledge workers supply, what can be owned, and what must be bargained over. My ongoing agenda links these tradeoffs to study when better information and new technologies translate into higher productivity, broader opportunity, and gains that workers share.